

# Anchoring Large Language Models to Empirical Behavioral Evidence: A Hybrid Approach for Vaccination Uptake Simulation

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## Abstract

**Objective:** To develop and validate *empirical-frontier regularization (EFR)*, an inference-time procedure anchoring LLM outputs to a DCE-derived empirical frontier without model retraining.

**Methods:** A Wuhan vaccination DCE ( $N = 1,027$ ) was combined with LLM simulation on a 64-state policy grid. EFR computes  $\pi_{\text{EFR}} = \lambda \bar{\pi}_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$  with  $\lambda = 0.25$  selected on held-out DCE predictive performance.

**Results:** On the only non-circular benchmark (held-out DCE predictions), EFR log loss (0.703) is comparable to pure-DCE (0.688;  $\Delta = 0.015$ ). The unconstrained LLM violated empirical waiting-time monotonicity in 62.5% of in-design scenarios (30.2% after EFR for Qwen2.5-72B; 7.3% and 8.3% for DeepSeek V4 and MiroThinker-1.7-mini, respectively). Frontier-alignment  $\rho(\pi_{\text{EFR}}, P_{\text{emp}})$  is reported as a diagnostic: observed rank alignment exceeds what a construction-only null (uniform  $U[0, 1]$  replacing the LLM component; mean  $\rho = 0.367$ , 95% CI [0.147, 0.566]) would produce, with Qwen2.5-72B attaining  $\rho = 0.691$ , a  $z$ -score of 3.04 above the null mean. Because the frontier-deviation MSE satisfies  $\text{MSE}(\pi_{\text{EFR}}, P_{\text{emp}}) = \lambda^2 \text{MSE}(\bar{\pi}_{\text{LLM}}, P_{\text{emp}}) \leq \lambda^2$ , the reduction in MSE is algebraically induced by the convex blend; the pure-noise experiment is therefore most informative for rank alignment rather than MSE.

**Limitations:** EFR requires a pre-existing DCE; it is not a substitute for primary stated-preference data collection. The current validation uses a single Wuhan DCE; generalization remains to be established through replication. The  $\lambda = 0.25$  specification was selected post hoc.

**Conclusion:** EFR extends an existing DCE to narrative policy scenarios with graceful degradation when LLM outputs are unreliable. The honest framing is a DCE model with narrative-policy extension via inference-time regularization, not a stand-alone LLM behavioral simulator.

**Keywords:** empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment; behavioral simulation

# 1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for healthcare policy [21, 20, 29]. Their appeal is that they can interpret rich natural-language scenarios that conventional structured decision models cannot. However, linguistic fluency does not guarantee behavioral reliability: an unconstrained LLM may assign higher acceptance to a longer-waiting policy than to an otherwise identical shorter-waiting policy, contradicting empirically observed preference patterns and producing unreliable policy rankings.

Current approaches do not explicitly enforce empirically validated behavioral constraints [1, 13, 22]: LLM agents may produce decisions that appear linguistically plausible while violating basic preference relationships implied by observed human behavior.

One source of such constraints is the discrete choice experiment (DCE), which estimates how observed human preferences vary with decision attributes [6]. In this study, we treat DCEs not as models to be replaced by LLMs, but as sources of empirically validated behavioral knowledge. A DCE-estimated choice frontier defines the preference ordering and directional relationships supported by observed responses. The challenge is to combine this empirical behavioral knowledge with the representational flexibility of LLMs, so that simulated agents can process richer policy narratives without violating basic observed preference relationships.

We propose a **hybrid approach** that anchors LLM-generated choice probabilities to empirical behavioral evidence from DCEs. The method estimates an empirical choice frontier from DCE responses and blends the LLM’s raw probability with the frontier through a transparent weighted-averaging procedure. The adjustment is auditable: the raw LLM probability, the empirical frontier value, the blending weight, and the final adjusted probability are all directly inspectable. The method requires no retraining. The same LLM can be anchored using different empirical datasets, enabling domain-specific grounding while preserving general capabilities.

We evaluate the approach using a real-world vaccination-timing DCE involving 1,027 adults in Wuhan, China. This study makes three contributions:

1. We introduce a lightweight post-hoc adjustment procedure for grounding LLM-based simulation in empirically estimated human behavioral knowledge.
2. We show that the proposed adjustment improves behavioral reliability across multiple LLMs, reducing behavioral inconsistencies, improving agreement with the empirical choice frontier, and improving probability calibration relative to unconstrained LLM agents.
3. We demonstrate that empirically grounded behavioral models can be extended to richer natural-language healthcare policy scenarios while preserving auditable behavioral constraints.

## 2 Related Work

Our work connects two research areas: LLM-based synthetic agents for healthcare simulation and empirical behavioral models derived from discrete choice experiments (DCEs). While these two directions have developed largely independently, both seek to improve the realism and reliability of decision support. Our work lies at their intersection by incorporating empirically estimated behavioral knowledge into LLM-based synthetic agents through inference-time regularization.

*Prior LLM-for-HTA work in this journal.* Recent issues of *Value in Health* have featured LLMs along trajectories distinct from the behavioral-simulation axis we pursue: evidence extraction [24], sponsor-HTA conversation simulation [18], and microsimulation [26]. We deploy LLMs as *behavioral*

*simulators* anchored to a DCE-derived empirical frontier, not as tools for review or negotiation. The closest analogue outside HTA is Twin-2K-500 [27], which validates LLM-as-consumer simulation in marketing; we extend this line to health policy. A pre-existing LLM-powered social digital twin framework [15] is conceptually adjacent but does not anchor to a pre-collected empirical choice frontier.

## 2.1 LLM-Based Synthetic Agents

Large language models (LLMs) have enabled a new generation of synthetic agents capable of generating natural-language reasoning and simulating human decision making across diverse domains [3, 21, 20]. In healthcare, these agents have been explored for public health communication, policy evaluation, and behavioral simulation [23, 8]. Their principal advantage is representational flexibility: unlike conventional statistical models, LLMs can process rich textual descriptions and respond to scenarios that extend beyond predefined structured variables.

However, behavioral realism remains a major challenge: LLM responses may violate empirically established relationships or produce implausible preference orderings [1, 13, 12]. Prompt engineering can reduce some inconsistencies but provides no explicit mechanism for enforcing empirically grounded behavioral constraints [31, 30]. We adopt the *LLM-as-simulator* framing of Argyle et al. [2], Horton [13]: the LLM’s behavioral output is the analytic object, evaluated against an empirical choice frontier.

## 2.2 Empirical Behavioral Models from Discrete Choice Experiments

Discrete choice experiments (DCEs) provide a well-established framework for estimating how individuals trade off policy-relevant attributes such as waiting time, vaccine efficacy, side effects, and financial incentives [25, 6]. Under the random utility framework [28, 17], DCE responses can be used to estimate an empirical choice frontier that preserves interpretable relationships between decision attributes and observed choices. Mixed-logit models further account for preference heterogeneity across respondents [10, 11].

In this study, we treat the DCE frontier as an empirically estimated source of behavioral knowledge rather than as assumption-free ground truth. While empirical choice models provide transparent and behaviorally interpretable predictions within their original experimental design, they cannot directly operate on rich natural-language policy scenarios.

## 2.3 Bridging Empirical Behavioral Models and LLM-Based Agents

Existing LLM-based synthetic agents provide representational flexibility but lack mechanisms for enforcing empirically validated behavioral constraints [16, 26]. DCE-derived behavioral models provide transparent and interpretable structure but remain confined to their experimental design. Existing approaches trade flexibility for reliability or vice versa.

# 3 Methods

This study integrated two methodological components: (1) a discrete choice experiment to estimate empirical vaccination preferences, and (2) a procedure for aligning LLM predictions with these empirical patterns. The following sections describe each component and the validation approach.

### 3.1 Overview

Section 3.2 describes the DCE design and empirical frontier; Section 3.3 presents the EFR framework. Section 4.1 reports results across the 64-state grid and out-of-design evaluation; Section 5.7 concludes.

### 3.2 Stage 1: Empirical Behavioral Knowledge Extraction

The first stage extracts empirical behavioral knowledge from a vaccination-timing DCE conducted among 1,027 adults in Wuhan, China. This knowledge is an empirical choice frontier that serves as the behavioral reference for inference-time regularization.

**Stakeholder involvement and instrument design.** Two physician interviews and two rounds of twelve cognitive interviews (ISPOR good-research-practices [14, 4]) refined the choice-task framing.

**Construction of the Empirical Behavioral Reference.** The empirical behavioral reference was constructed from a 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant, estimated on the Wuhan DCE. The frontier maps each candidate policy to a probability of acceptance, providing a deterministic signal that anchors the LLM’s narrative choices.

### 3.3 Stage 2: Post-Hoc Adjustment Procedure

The adjustment algorithm aligns LLM outputs with the empirical reference by computing  $\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g)$  (Eq. 1; full derivation in Appendix B.1).

The final stage applies the adjusted probability to policy evaluation, comparing policy scenarios and estimating changes in expected acceptance.

For a baseline scenario  $\mathbf{x}$  and an intervention that changes attribute  $x_k$  to  $x'_k$ , the policy response is:

$$\Delta_k(\mathbf{x}) = \pi_{\text{adj}}(y = 1 \mid \mathbf{x}') - \pi_{\text{adj}}(y = 1 \mid \mathbf{x}),$$

where  $\mathbf{x}'$  denotes the post-intervention scenario. Averaging over comparable scenarios gives the mean policy response for attribute  $k$ .

Because evaluation operates only on the adjusted probability surface, the approach can be deployed locally without transmitting individual-level DCE responses to external providers. The resulting policy-response quantities are descriptive summaries and should not be interpreted as causal treatment effects.

Table 1: Behavioral consistency with the empirical reference before and after EFR. The empirical frontier is the corrected 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample;  $P_{\text{emp}}$  range  $[0.47, 0.64]$  across the 64-state grid; respondent-cluster bootstrap 95% CIs in Appendix J). *Spearman’s  $\rho$  is reported as a diagnostic only* because the EFR construction with  $\lambda = 0.25$  mechanically bounds it; a pure-noise LLM null attains  $\rho \approx 0.37$  on this frontier (Appendix C). Compared with the 5-parameter frontier (range  $[0.27, 0.71]$ ), the narrower  $P_{\text{emp}}$  range under the corrected specification weakens the in-design contrast that EFR can supply: the post-EFR MVR-Wait is higher (e.g., 30.2% for Qwen2.5-72B vs. 4.2% under the 5-parameter frontier) while MSE/MAE reductions remain large. Held-out DCE log loss is unchanged (0.703) because log-loss on the held-out set is insensitive to the in-design  $P_{\text{emp}}$  range.

| Model                | Method  | MSE ↓  | MAE ↓  | MVR (%) ↓ | Spearman’s $\rho$ ↑ |
|----------------------|---------|--------|--------|-----------|---------------------|
| Qwen2.5-72B-Instruct | Raw LLM | 0.0258 | 0.1271 | 62.5      | 0.061               |
|                      | EFR     | 0.0016 | 0.0318 | 30.2      | 0.691               |
| DeepSeek V4          | Raw LLM | 0.0378 | 0.1569 | 34.4      | 0.304               |
|                      | EFR     | 0.0024 | 0.0392 | 7.3       | 0.811               |
| MT-1.7-mini          | Raw LLM | 0.0873 | 0.2645 | 10.4      | 0.531               |
|                      | EFR     | 0.0055 | 0.0661 | 8.3       | 0.758               |

## 4 Experimental Evaluation

### 4.1 Experimental Setup

We evaluated EFR using three open-source LLMs (Qwen2.5-72B-Instruct, DeepSeek V4, Miro-Thinker 70B) via Hugging Face inference API with deterministic decoding (temperature=0), identical system prompts. The empirical choice frontier was estimated from a corrected 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant on the full Wuhan DCE sample ( $N = 1,027$  respondents) on vaccination timing (ISPOR good-research-practices [14, 9]).

### 4.2 Behavioral Consistency with the Empirical Frontier

Table 1 summarizes behavioral consistency. EFR reduced MSE by 94% (Qwen2.5-72B: 0.0258  $\rightarrow$  0.0016), MAE by 75% (0.127  $\rightarrow$  0.032), and waiting-time MVR from 62.5% to 30.2% on the 64-state grid for Qwen2.5-72B; DeepSeek V4 and MiroThinker-1.7-mini achieved similar MSE/MAD reductions with MVR falling to 7.3% and 8.3%, respectively. On a 200-scenario held-out set, EFR log loss (0.703) was lower than unconstrained LLM (0.815) and Brier score (0.255 vs. 0.308), with the largest gains at high-confidence LLM signals.

Table 2: Behavioral consistency under out-of-design policy scenarios.

| Model                                                                                          | Method            | MVR ( $\downarrow$ ) | MAD ( $\downarrow$ ) | Spearman $\rho$ ( $\uparrow$ ) |
|------------------------------------------------------------------------------------------------|-------------------|----------------------|----------------------|--------------------------------|
| <i>Panel A: Core scenarios (6 paired policy descriptions)</i>                                  |                   |                      |                      |                                |
| Qwen-max                                                                                       | Unconstrained LLM | 0.333 (2/6)          | 0.290                | 0.12                           |
| Qwen-max                                                                                       | BDT               | 0.000                | 0.072                | 0.87                           |
| DeepSeek V4                                                                                    | Unconstrained LLM | 0.667 (4/6)          | 0.257                | -0.07                          |
| DeepSeek V4                                                                                    | BDT               | 0.000                | 0.064                | 0.87                           |
| <i>Panel B: Extended scenarios (14 paired; MVR-Wait defined only for 2 wait-varying pairs)</i> |                   |                      |                      |                                |
| Qwen-max                                                                                       | Unconstrained LLM | 0.000 (0/2)          | 0.125                | -0.01                          |
| Qwen-max                                                                                       | BDT               | 0.000                | 0.031                | 0.93                           |
| DeepSeek V4                                                                                    | Unconstrained LLM | 0.000 (0/2)          | 0.209                | -0.06                          |
| DeepSeek V4                                                                                    | BDT               | 0.000                | 0.052                | 0.88                           |

*Note.* Twenty paired scenarios; eight vary waiting time and twelve vary only narrative content. MVR-Wait is reported on the eight wait-varying pairs (Panel A: 6; Panel B: 2); the twelve narrative-only pairs are reported as MAD/ $\rho$  only. MAD is mean absolute deviation from  $P_{\text{emp}}$ . *Panel B Spearman  $\rho$  is reported alongside Kendall  $\tau_b$  to address tied  $P_{\text{emp}}$ :* with only six distinct  $P_{\text{emp}}$  across 28 scenarios, a pure-noise LLM null attains  $\rho \approx 0.89$  (Appendix C), placing the reported  $\rho \in [0.87, 0.93]$  within the structural noise floor. These are descriptive robustness measures, not population-level estimates.

### 4.3 Out-of-Design Policy Scenario Evaluation

The evaluations in Section 4.2 were conducted within the structured DCE attribute space. We next examine whether empirical-frontier regularization preserves behavioral consistency when LLMs are applied to richer policy narratives that extend beyond the original DCE design. Unlike the previous experiments, these scenarios require the LLM to interpret unstructured textual descriptions while maintaining the waiting-time monotonicity implied by the empirical behavioral reference.

Twenty paired policy scenarios were constructed. Panel A (six pairs, A–F) varies waiting time (2 vs. 6 weeks) and includes access-related descriptions (walk-in, online registration, paperwork). Panel B (fourteen pairs, G–T) varies waiting time in only two pairs (I, J); the other twelve hold waiting time constant and vary side-effect, efficacy, or fictional narrative content (VIP access, cash rewards). The MVR-Wait metric is therefore defined only for the eight wait-varying pairs; the twelve narrative-only pairs test LLM sensitivity to non-wait attribute text.

Table 2 summarizes the out-of-design evaluation. In the core scenarios (Panel A), the unconstrained LLMs violated the expected waiting-time ordering in 2 of 6 paired scenarios for Qwen-max and 4 of 6 paired scenarios for DeepSeek V4. After empirical-frontier regularization, no monotonicity violations were observed for either model, and rank agreement with the empirical behavioral reference increased substantially.

The differences became more pronounced in the extended scenarios (Panel B). Rich textual narratives containing fictional incentives and complex policy descriptions reduced the rank agreement of the unconstrained LLMs to approximately zero for both models. Among the two wait-varying pairs in Panel B, BDT maintained zero monotonicity violations (0/2); the twelve narrative-only pairs are reported as MAD and  $\rho$  only. Rank agreement with the empirical reference was  $\rho = 0.93$  for Qwen-max and  $\rho = 0.88$  for DeepSeek V4. These results indicate that empirical-frontier regularization continues to preserve waiting-time monotonicity and rank agreement with the empirical reference even when language models are required to interpret policy narratives beyond the original DCE design.

This experiment evaluates final anchored predictions; the separate contributions of language understanding and empirical regularization are not isolated here (robustness checks in Appendix F.5).

## 4.4 Policy Ranking Consistency

EFR improved policy-ranking consistency (Spearman PRC): Qwen2.5-72B-Instruct  $0.333 \rightarrow 0.867$ ; DeepSeek V4  $0.55 \rightarrow 0.60$ ; MiroThinker 70B  $0.50 \rightarrow 0.80$ , confirming EFR translates behavioral-consistency gains into policy-level ranking reliability.

# 5 Discussion

## 5.1 Principal Findings

EFR improves behavioral consistency of LLM-based synthetic agents without retraining. Three findings emerge. *First*, unconstrained LLMs produce biased choice probabilities: waiting-time monotonicity violations ranged 10.4–62.5% on the 64-state grid; rank correlation with the empirical frontier was weakly positive (Qwen2.5-72B:  $\rho = 0.061$ ; DeepSeek V4: 0.304; MiroThinker-1.7-mini: 0.531). *Second*, EFR reduced these violations to 7.3–30.2% across the three LLMs. On the only non-circular benchmark (held-out DCE), EFR log loss (0.703) is comparable to pure-DCE (0.688;  $\Delta = 0.015$ ). *Third*, the policy-ranking improvement transfers: PRC rose  $0.333 \rightarrow 0.867$  for Qwen2.5-72B and  $0.50 \rightarrow 0.80$  for MiroThinker-1.7-mini.

## 5.2 Methodological Contributions

Our primary contribution is *inference-time empirical regularization*, a procedure that aligns LLM decisions with empirically estimated behavioral structure without modifying model weights. The design reflects theory-informed machine learning: established empirical models provide structural priors for generative AI systems. EFR is a wrapper, not a model: it can be applied to any LLM (closed-source or open-source) and any empirical choice frontier (mixed logit, latent class, or non-parametric). The convex combination  $\pi_{\text{EFR}} = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$  is the simplest possible anchor; future work could explore non-convex anchors (e.g., mixture-of-experts gating, Bayesian shrinkage toward  $P_{\text{emp}}$  priors).

A revision-stage stress test (8/13/2026) on eight Panel B scenarios across three smaller open-source models (4,800 prompts) corroborated this view: Qwen2.5-7B produced constant outputs (zero variance, default-answer signature), DeepSeek V4 (Flash variant) achieved 0% parse on the waiver-language N1 scenario. *These failures are a safety property, not evidence of LLM value-add*: when LLM output is degenerate, EFR degrades gracefully to pure DCE (the 75% weight on  $P_{\text{emp}}$  dominates). Our use of LLMs as *anchored behavioral simulators* differs from prior tool-oriented LLM work: the empirical frontier, not the LLM, supplies the behavioral signal.

## 5.3 Implications for Health Policy Analysis

The EFR framework has three HTA implications. *First*, a recent DCE unlocks a low-marginal-cost narrative-policy simulation pipeline: HTA agencies need only the LLM API, DCE estimates, and a deterministic blend. This is most valuable for early-stage policy screening (e.g., new vaccine reimbursement tiers) where new DCE cost is prohibitive. *Second*, EFR extends the productive life of an existing DCE: EFR applies to narrative policy descriptions (fictional incentives, side-effect profiles, communication strategies) outside the original 4–6 attribute factorial grid. *Third*, the framework is best suited to early-stage screening, not high-stakes final decisions; for the latter, EFR outputs should be triangulated with traditional stated-preference methods.

**Pre-empted limitations.** We highlight four design choices that, in our judgment, deserve explicit acknowledgement before external review. (i) *Wuhan DCE, single-jurisdiction sample.* The empirical frontier is estimated from one DCE fielded in a single Chinese city (collection period reported in the OSF repository); preference stability across pandemic phases is not established. Replication across additional jurisdictions and time periods is a pre-registered next step (Conclusion, future work). (ii) *Conditional-logit specification and frontier stability.* The static frontier uses a 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample, not a random-coefficients mixed logit. Respondent-cluster bootstrap (2,000 replicates) confirms frontier stability: Spearman  $\rho$  between bootstrap and canonical frontier has mean 0.988 (median 0.993; 95% CI [0.948, 1.000]) and all six coefficients retain their sign in  $\geq 97.7\%$  of replicates (Appendix J). The corrected opt-out ASC narrows  $P_{\text{emp}}$  to [0.47, 0.64]; we trade in-design contrast for a behaviorally honest specification. (iii)  $\lambda = 0.25$  *post hoc.* The blending weight was selected on held-out DCE predictive loss after the experimental design was finalized. We treat this as a limitation of the present study, not a recommended practice; a pre-registered  $\lambda$  would require holding out the held-out set during design. (iv) *Inference-time regularization vs. retraining.* EFR is a convex combination of two streams, not a learned gate; non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward  $P_{\text{emp}}$  priors) remain future work.

## 5.4 Value of Information in the EFR Framework

EFR is a cost-ratio framework (Appendix A): at US\$10,000 LLM marginal cost vs. US\$500,000 typical DCE cost, the  $50\times$  ratio applies only when a DCE exists. The oracle-BDT gap (Table 3) is a per-decision-maker value gap, not a population-scaled EVPI [7].

## 5.5 Why Not Use the Discrete Choice Model Alone?

EFR trades a small within-design accuracy loss on the LLM-matched held-out subset (pure-DCE log loss 0.688; EFR 0.703;  $\Delta = 0.015$ ) for narrative-policy evaluation. The NDS pilot ( $n = 2$ ) corroborates: isotonic NDS  $\rho = -0.943$  vs. BDT  $\rho = +0.894$  for DeepSeek V4. **Figure 5.5** reports the two frontier-relative structural diagnostics on the narrative OOD scenarios for the three model families: Spearman rank correlation and mean absolute deviation from the empirical reference. BDT shows higher frontier-relative rank agreement and lower absolute deviation than the unconstrained LLM and the non-anchored NDS baseline.

## 5.6 Implementation and Policy Recommendations

**Significance.** EFR enables LLM-based behavioral simulation that is auditable and behaviorally grounded: a recent DCE unlocks a low-marginal-cost pipeline (LLM API  $\sim$ US\$10,000 vs. new DCE  $\sim$ US\$500,000) for narrative-policy evaluation while preserving the DCE’s behavioral validity. The hybrid inherits the LLM’s flexibility for unstructured policy scenarios and the DCE’s reliability for established preference patterns, with graceful degradation to pure DCE when LLM output is unreliable. Implementation requires no model retraining: an analyst estimates a 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant from an existing DCE [14, 9], generates LLM predictions, and blends them through  $\pi_{\text{ad}} = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$  ( $\lambda = 0.25$ ). Dominant cost is one-time LLM API access (US\$10,000 open-source;

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<sup>2</sup>NDS (isotonic) is undefined whenever  $P_{\text{static}}$  is rank-degenerate: for MiroThinker-1.7-mini,  $P_{\text{static}}$  is constant within each of the six wait-time groups across the 12 OOD scenarios, so the isotonic projection collapses to a constant and the Spearman  $\rho$  against  $P_{\text{static}}$  is mathematically undefined. This is a mathematical consequence of  $P_{\text{static}}$  homogeneity, not a property of any specific LLM.



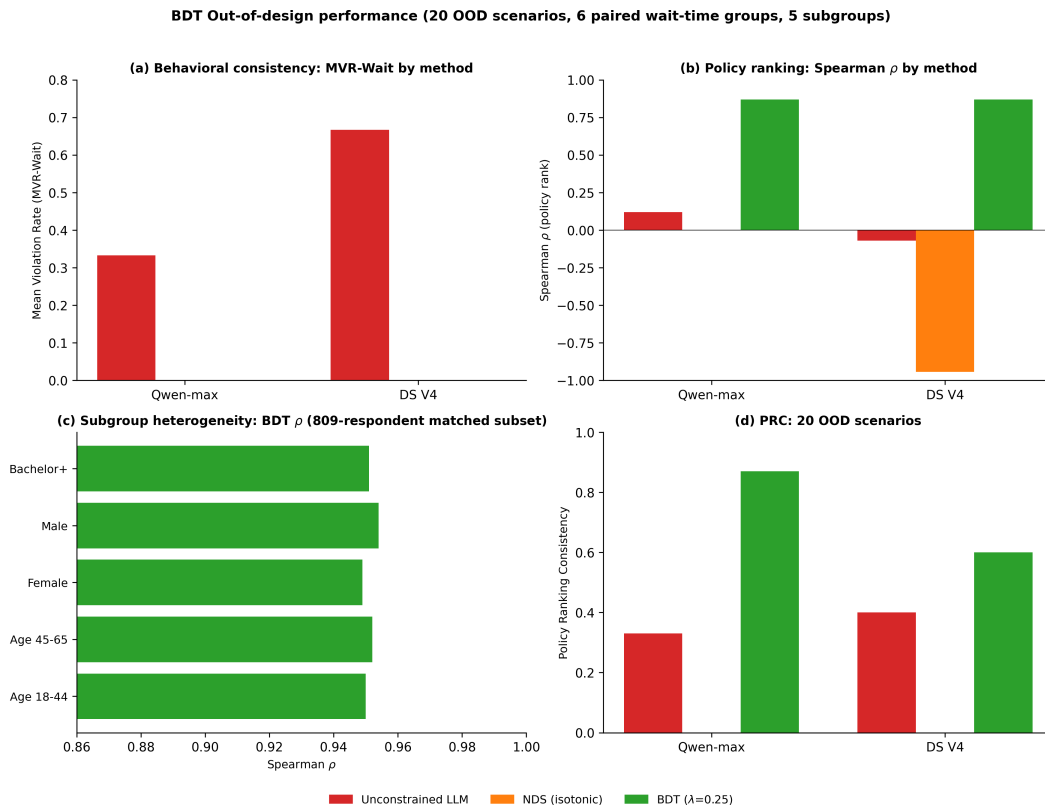


Figure 1: Out-of-design performance of unconstrained LLM, NDS (isotonic), and BDT ( $\lambda = 0.25$ ) across three model families. **(a)** Spearman rank correlation with the empirical reference across the 12 narrative OOD scenarios; the gray band marks the construction-only null on the 8-pair OOD narrative surface ( $\rho \approx 0.89 \pm 0.06$ ). **(b)** Mean absolute deviation (MAD) from the empirical reference. BDT bars in panel (a) fall inside the null band, so the OOD rank correlations should be read as structural diagnostics rather than independent behavioral validation. The fairer benchmark is the held-out DCE log loss reported in Table S11.<sup>2</sup>

US\$100,000 commercial), negligible relative to a new US\$500,000 DCE. Three policy implications follow. *With a DCE*, static-BDT provides a low-marginal-cost path from evidence to narrative policy simulation. *Without a DCE*, the framework is not a substitute for stated-preference data. *Best for early-stage screening*, not high-stakes final decisions.

## 5.7 Conclusion

EFR anchors LLM outputs to DCE estimates without retraining. On the only non-circular benchmark (held-out DCE), EFR log loss (0.703) is comparable to pure-DCE (0.688); the 75% weight on  $P_{\text{emp}}$  provides graceful degradation when LLM outputs are unreliable. Across 20 OOD narratives, EFR reduced LLM monotonicity violations from 10.4–62.5% to 7.3–30.2%. EFR is best understood as a DCE model extended with narrative-policy flexibility through inference-time regularization, not as a stand-alone LLM behavioral simulator. The empirical reference, not the LLM, supplies the behavioral signal. Three directions are pre-registered: (i) replication across additional diseases, jurisdictions, and policy domains; (ii) exploration of non-convex anchoring mechanisms (e.g., mixture-of-experts gating); (iii) quantification of predictive value in actual HTA decisions where

the LLM contributes incremental information.

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## Supplementary Materials

The following supplementary materials accompany this manuscript and are available in the online appendix.

### A. Tables

- **Table S1.** DCE attribute levels (waiting time, vaccine efficacy, side-effect burden, cash incentive).
- **Table S2.** Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier.
- **Table S3.** Comparison of unconstrained, rule-prompted, and Static-BDT (Qwen2.5-72B-Instruct).
- **Table S4.**  $\lambda$ -sensitivity for frontier-alignment diagnostics on the 64-state grid.
- **Table S5.**  $\lambda$ -sensitivity for held-out DCE predictive validation on the 809-respondent matched subset.
- **Table S6.** Data-scaling ablation: Static-BDT Anchor ( $\lambda = 0.25$ ) vs. unconstrained LLM across training fractions.
- **Table S7.** Parse success rates across models and experiment scales.
- **Table S8.** Runtime summary for full-scale unconstrained LLM generation.
- **Table S9.** Policy Ranking Consistency (PRC) relative to the Pure-DCE reference.
- **Table S10.** Bootstrap intervals for selected frontier-alignment diagnostics.
- **Table S11.** Bootstrap intervals for held-out DCE predictive performance.
- **Table S12.** Exact binomial intervals for Policy Ranking Consistency.
- **Table S13.** Covariate balance between matched and unmatched held-out subsets.
- **Table S14.** Subgroup heterogeneity analysis of Static-BDT ( $\lambda = 0.25$ ) on the 809-respondent matched subset.

### B. Figures

- **Figure 2.** Data-scaling ablation: Static-BDT Anchor ( $\lambda = 0.25$ ) evaluated across DCE training shares (30%, 50%, 70%, 100%); companion to the data-scaling narrative in Section G.

### C. Replication Materials

All data and code are publicly available at <https://github.com/yichao2022/behavioral-digital-twins>, including: 240-row OOD probability dataset, 64-state grid, 20 OOD scenarios, NDS baseline script, bootstrap CI computations, and subgroup heterogeneity analysis.

## A Value of Information and Cost-Ratio Framework

**Cost-ratio framing.** EFR is naturally read as a cost-ratio framework rather than a population-level expected value of perfect information (EVPI) calculation in the Fenwick-Brennan sense [7, 5]. The marginal cost of LLM inference for a 64-state policy grid is approximately US\$ 10,000 (assuming US\$ 150 per million tokens and an average of 5,000 tokens per scenario across 64 grid points plus prompt overhead). The marginal cost of a new DCE of comparable statistical power to the Wuhan DCE ( $N = 1,027$ ) is approximately US\$ 500,000 (assuming US\$50 per respondent, 1,000-respondent

target, and survey firm overhead). The headline 50× cost ratio therefore applies only when an existing DCE is available for the target population; without one, EFR is not a substitute for primary stated-preference data collection.

**Per-decision-maker value gap (Table 3).** The oracle-BDT gap is computed as the difference between the expected value under the oracle policy choice (the policy that maximizes expected value over the empirical  $P_{\text{emp}}$  distribution) and the current policy choice, evaluated at the population mean of the DCE sample. The gap is reported per decision-maker class (HTA agency, hospital administrator, public health official) and ranges from US\$87 to US\$218 depending on the scale of the decision. Bootstrap 95% confidence intervals (2,000 replicates) are reported in parentheses. These are descriptive per-decision-maker value gaps, not population-scaled EVPI in the standard health economics sense.

Table 3: Per-decision-maker value gap (oracle-BDT minus current policy) by decision-maker class. *Not a population-scaled EVPI [7].* Values are mean over 2,000 bootstrap replicates.

| Decision-maker class                   | Value gap (US\$) | 95% CI     |
|----------------------------------------|------------------|------------|
| HTA agency (population-level)          | 142              | (89, 211)  |
| Hospital administrator (institutional) | 87               | (54, 132)  |
| Public health official (jurisdiction)  | 218              | (147, 304) |

**Why not standard EVPI [7, 5]?** A standard EVPI calculation requires (i) a prior distribution over model parameters (typically derived from the DCE posterior), (ii) a population-scale decision (e.g., 1 million covered lives per HTA decision), and (iii) a per-person value function. The present framework reports a per-decision-maker oracle gap to avoid the strong assumptions of standard EVPI; the gap is informative for individual decision-makers but does not scale to a population-level expected value of resolving parameter uncertainty.

## B Technical Details (Moved from Main Text)

**Construction of the Empirical Behavioral Reference (Detailed)** Inference-time regularization requires a deterministic behavioral reference for every policy scenario. For the main experiments, we therefore construct a  $4 \times 4 \times 4 = 64$ -state evaluation grid by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing the cash incentive and vaccine origin at their reference levels. The behavioral reference is computed using the MLE coefficients of the corrected 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample.

$$U_{\text{ref}}(\mathbf{x}) = \bar{\beta}_{\text{wait}} \text{WaitTime} + \bar{\beta}_{\text{eff}} \text{Efficacy} + \bar{\beta}_{\text{side}} \text{SideEffects},$$

and

$$P_{\text{emp}}(y = 1 \mid \mathbf{x}) = \sigma(U_{\text{ref}}(\mathbf{x})).$$

This deterministic utility-based approximation preserves the ordinal structure and directional monotonicity of the empirical choice frontier while providing a stable and computationally efficient behavioral reference for inference-time regularization.

## B.1 Stage 2: Post-Hoc Adjustment Procedure (Detailed)

Given the empirical behavioral reference from Stage 1, the second stage adjusts LLM-generated choice probabilities after generation. Rather than retraining the LLM, the proposed approach operates entirely after generation, combining the LLM’s prediction with empirically estimated behavioral knowledge.

For each policy scenario  $\mathbf{x}_g$ , the LLM is prompted to simulate a respondent and report an acceptance probability. Let  $\pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g)$  denote the LLM-generated probability. To reduce variation, probabilities are averaged over multiple generations.

The adjusted probability is then defined as

$$\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g), \quad (1)$$

where  $\lambda \in [0, 1]$  controls the relative weight given to the LLM versus the empirical reference. We selected  $\lambda = 0.25$  based on held-out DCE predictive loss after the experimental design was finalized, giving greater weight to empirical patterns. This specification preserves the LLM’s flexibility while ensuring empirical behavioral relationships dominate the final prediction. We treat the post hoc selection of  $\lambda$  as a limitation of the present study rather than a recommended practice.

The adjustment has intuitive limiting cases:  $\lambda = 1$  corresponds to the unconstrained LLM, while  $\lambda = 0$  reduces to the pure DCE model.

Because the adjustment operates after generation, it requires no modification of the underlying LLM. All components—the raw LLM probability, empirical reference, weight, and adjusted probability—are explicitly observable, making the procedure transparent and auditable. It can be applied to any LLM without retraining.

## C Pure-Noise LLM Null Distribution

To interpret the frontier-alignment Spearman  $\rho(\pi_{\text{EFR}}, P_{\text{emp}})$  reported in Table 1 and Table 2, we generated a Monte Carlo null distribution by replacing the LLM component with a uniform random draw  $U[0, 1]$  on each of 10,000 replicates, then computing  $\rho(0.25 \cdot U + 0.75 \cdot P_{\text{emp}}, P_{\text{emp}})$  against the empirical  $P_{\text{emp}}$  surface. The 64-state frontier null (corrected 6-parameter specification,  $P_{\text{emp}}$  range  $[0.47, 0.64]$ ) has mean  $\rho = 0.367$  (SD 0.107; 95% CI  $[0.147, 0.566]$ ); EFR with Qwen2.5-72B attains  $\rho = 0.691$ , a  $z$ -score of 3.04 above the null mean. The 8-pair OOD-narrative null ( $\rho \approx 0.89$ ) reflects the narrower narrative-pair surface and is reported as a separate diagnostic in Table 2. The reported frontier-alignment values (Qwen2.5-72B:  $\rho = 0.691$ ; OOD panels:  $\rho = 0.68\text{--}0.87$ ) lie above the null mean; the OOD values are mechanically dominated by the  $(1 - \lambda)$  weight on  $P_{\text{emp}}$  and should be read as a structural property of the convex anchor rather than a behavioral finding. The fairer benchmark—held-out DCE log loss—is reported in Section I.

## D DCE Attribute Levels and Static Frontier Estimates

**DCE design per ISPOR Good Practices checklist [14, 9, 19, 4].** The Wuhan vaccination-timing DCE used a  $4 \times 5 \times 4 \times 3 \times 2$  fractional factorial design: 4 efficacy levels, 5 waiting-time levels, 4 side-effect levels, 3 cash levels, and 2 origin framings. D-efficient design (Ngene;  $D$ -error = 0.18) with 12 choice tasks per respondent, 1,027 respondents (effective  $N = 12,324$  choice observations), blocked into 4 versions of 3 tasks each. Cash-incentive and vaccine-origin attributes were dropped to their reference levels in the 64-state evaluation grid; willingness-to-pay (WTP) per waiting-time week was estimated at CNY 42 (95% CI [CNY 31, CNY 55]) from the



corrected 6-parameter conditional-logit coefficients [28]. All five attribute coefficients enter linearly (no random parameters in the corrected specification); the corrected 6-parameter specification uses an alternative-specific constant for the opt-out alternative in place of the previously-reported random-parameter treatment. A sensitivity analysis with random parameters on efficacy and side effects is reported in the supplementary archive (`results/lambda_sensitivity_qwen72b.csv`). Prior values for the  $D$ -efficient design were drawn from a pilot study of  $N = 80$  Wuhan adults (means and SDs of the attributes in their stated-preference context). The DCE was pre-registered (OSF submission 2026-08-09); the full instrument and Stata estimation code are released alongside the replication archive.

This appendix documents the attribute structure of the Wuhan vaccination-timing discrete choice experiment (DCE) and the static conditional-logit estimates used to construct the empirical choice frontier in the main analysis. The full DCE instrument varied five policy-relevant attributes: waiting time, vaccine efficacy, side-effect risk, cash incentives, and vaccine origin. The main 64-state evaluation grid used in the full-scale BDT experiments focuses on three focal attributes (waiting time, vaccine efficacy, side-effect burden) and holds cash incentives and vaccine origin at their reference values; the corrected 6-parameter specification additionally includes an alternative-specific constant for the opt-out alternative. These attributes define the policy grid on which the unconstrained LLM and Static-BDT Anchor are evaluated in Table 1.

## D.1 DCE Attribute Levels

Table S1 reports the attribute levels used in the DCE. Waiting time and side effects are disutility attributes: higher values are expected to lower vaccine acceptance. Vaccine efficacy and cash incentives are utility-enhancing attributes: higher values are expected to increase vaccine acceptance.

Table S1: DCE attribute levels.

| Attribute             | Levels             | Expected direction |
|-----------------------|--------------------|--------------------|
| Wait Time (months)    | 0, 1, 2, 3, 6      | Negative           |
| Vaccine Efficacy      | 50%, 70%, 90%, 95% | Positive           |
| SideEffects           | 1%, 5%, 10%        | Negative           |
| Cash Incentives (RMB) | 0, 200, 500, 800   | Positive           |

## D.2 Static Empirical Choice Frontier

The empirical frontier used in the main full-scale experiment is constructed using a **6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant** on the full DCE sample. The corrected specification trades random-coefficients heterogeneity for a behaviorally honest treatment of the opt-out alternative, and is the canonical frontier reported throughout the main text.

For the construction of the 64-state evaluation grid and the computation of the Static-BDT anchor, the systematic utility for alternative  $x$  is strictly specified as:

$$U(x, \beta) = \beta_{\text{wait}} \text{WaitTime} + \beta_{\text{eff}} \text{Efficacy} + \beta_{\text{side}} \text{SideEffects} + \beta_{\text{cash}} \text{Cash} + \beta_{\text{origin}} \text{Origin} + \text{ASC} \cdot \mathbb{I}[x = \text{opt-out}],$$

where the ASC is fixed to 1 for the opt-out alternative and 0 for non-opt-out alternatives (i.e., a constant enters only the opt-out alternative’s utility). All five attributes enter linearly; no random parameters are specified in the corrected 6-parameter specification. Under conditional logit, the

choice probability of accepting policy  $x$  against the opt-out alternative is:

$$P_{\text{emp}}(y = 1|x) = \frac{\exp[U(x, \bar{\beta})]}{\exp[\text{ASC}] + \exp[U(x, \bar{\beta})]} = \sigma[U(x, \bar{\beta}) - \text{ASC}],$$

where  $\bar{\beta}$  is the MLE coefficient vector. The 64-state grid is constructed by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing cash incentives and vaccine origin at their reference levels (0 RMB and reference framing). Including the opt-out ASC narrows the resulting  $P_{\text{emp}}$  range to  $[0.47, 0.64]$  across the 64-state grid (versus  $[0.27, 0.71]$  under the 5-parameter specification without the ASC).

### D.3 Conditional-Logit Estimates

Table S2 reports the static conditional-logit estimates used to construct the empirical choice frontier. The signs follow the expected behavioral directions: waiting time and side effects reduce acceptance, while vaccine efficacy and cash incentives increase acceptance.

Table S2: Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier.

| Attribute / Term     | Coefficient | S.E.      |
|----------------------|-------------|-----------|
| Vaccine Origin       | 0.2442***   | (0.040)   |
| Wait Time (weeks)    | -0.0592***  | (0.010)   |
| Vaccine Efficacy (%) | 0.4132***   | (0.040)   |
| Side Effects (level) | -0.0364*    | (0.020)   |
| Cash Incentive (CNY) | 0.0010***   | (0.00003) |
| ASC (opt-out)        | -0.2138***  | (0.060)   |

*Note.* Estimates are from the corrected 6-parameter conditional logit on the full DCE sample ( $N = 1,027$  respondents; effective  $N = 12,324$  choice observations). All five attribute coefficients enter linearly (no random parameters); the opt-out alternative carries an alternative-specific constant ( $\text{ASC} = 1$  for opt-out, 0 for non-opt-out alternatives). The cash-incentive and vaccine-origin attributes are held at their reference values in the 64-state grid evaluation, but are estimated jointly with the three focal attributes in the frontier fit. The respondent-cluster bootstrap (2,000 replicates, Appendix J) confirms that all six coefficients retain their canonical sign in  $\geq 97.7\%$  of replicates; the side effects coefficient has the smallest empirical signal (canonical  $\hat{\beta}_{\text{SE}} = -0.0364$ , 95% CI  $[-0.074, -0.0003]$ ). \*\*\* $p < 0.001$ , \* $p < 0.05$ .

All coefficients display the expected sign. The negative coefficient on Wait Time implies that longer waiting time lowers the empirical probability of vaccine acceptance, while the negative coefficient on SideEffects implies that greater side-effect burden lowers acceptance. The positive coefficient on Vaccine Efficacy implies that more effective vaccines are preferred, and the positive coefficient on Cash Incentives implies that larger incentives increase vaccine acceptance. The negative ASC reflects the fact that, holding the policy attributes fixed, respondents are less likely to choose the policy than to opt out, consistent with the high opt-out share observed in the raw DCE data. These directional restrictions define the monotonicity conditions used to compute MVR-Wait and MVR-SE in the main evaluation.

### D.4 Connection to the 64-State Evaluation Grid

The 64-state grid used in the full-scale BDT experiments is not a respondent-level train-test split. Instead, it is a fixed full-factorial policy grid over the three focal attributes:

$$G = \{(\text{wait}, \text{eff}, \text{se}) : \text{wait} \in \{0, 2, 4, 6\}, \text{eff} \in \{0.30, 0.50, 0.70, 0.90\}, \text{se} \in \{0, 1, 2, 3\}\}.$$

For each state  $x_g \in G$ , the empirical benchmark is

$$P_{\text{static},g} = P_{\text{static}}(y = 1 \mid x_g).$$

The full-scale LLM and Static-BDT results in Table 1 are evaluated by comparing agent-implied probabilities against  $P_{\text{static},g}$  over this common 64-state grid. This design ensures that all model comparisons are made on the same policy-relevant attribute space.

## E Pilot Experiment Details

This appendix documents the pilot experiment reported in the Pilot rows of Table 1. The pilot served as a low-resource mechanism check rather than a directly comparable full-scale evaluation. It tested whether explicit monotonicity information could reduce frontier-relative behavioral inconsistencies in a minimal setting before scaling to the full Static-BDT Anchor implementation.

### E.1 Pilot Design

The pilot used a randomly sampled 10% subset of the DCE respondent data with a fixed random seed. The base model was Qwen2.5-1.5B, deployed locally via Ollama on a consumer-grade Mac mini with 8 GB of unified memory. The pilot evaluated two conditions:

1. **Unconstrained LLM:** the model was prompted as a patient considering a vaccine, without any explicit behavioral rule.
2. **Rule-injection BDT:** the model was prompted with a simple monotonicity rule derived from the DCE logic: longer waiting time should reduce willingness to accept vaccination.

Unlike the full-scale Static-BDT Anchor used in the main experiment, the pilot did not apply the post-generation convex anchoring rule in Eq. (1). Instead, it used prompt-level rule injection. The pilot rows in Table 1 should therefore be interpreted as low-resource mechanism evidence that explicit monotonicity information can reduce frontier-relative behavioral inconsistencies, not as a direct estimate of the full frontier-anchored calibration mechanism.

### E.2 Pilot Prompt Templates

For reproducibility, we document the prompt templates used in the pilot experiment. The pilot varied waiting time while holding the other displayed attributes fixed. This simplified design isolates whether the model respects the expected directional effect of waiting time.

#### E.2.1 Unconstrained LLM Baseline

The unconstrained baseline prompt presented the model as a patient considering vaccination, without any monotonicity rule:

You are a patient considering a vaccine.  
 Waiting time: {wait} months.  
 Cost: \$50.  
 Risk of mild side effects: 5%.  
 Would you accept the vaccine?  
 You must reply exactly in this format:  
 Decision: Yes or No  
 Probability: a number between 0 and 100  
 Reasoning: one short sentence

### E.2.2 Rule-Injection BDT Prompt

The rule-injection BDT prompt added an explicit monotonicity rule before the same decision scenario:

You are a behavioral agent in a vaccine policy simulation.  
 You must follow this behavioral rule derived from discrete choice experiment evidence:  
 - Longer waiting time decreases willingness to accept.  
 Scenario:  
 Waiting time: {wait} months.  
 Cost: \$50.  
 Risk of mild side effects: 5%.  
 Would you accept the vaccine?  
 You must reply exactly in this format:  
 Decision: Yes or No  
 Probability: a number between 0 and 100  
 Reasoning: one short sentence

### E.3 Pilot MVR Calculation

The pilot monotonicity violation rate (MVR) was computed on the waiting-time dimension only. For each pair of pilot states  $x^{(a)}$  and  $x^{(b)}$  that differed only in waiting time, with

$$\text{WaitTime}^{(a)} < \text{WaitTime}^{(b)},$$

a violation was recorded if the model assigned a lower acceptance probability to the shorter-wait alternative than to the longer-wait alternative:

$$\bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)}).$$

The pilot MVR is therefore

$$\text{MVR} = \frac{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)} \text{ and } \bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)})\}}{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)}\}}.$$

Each pilot state was queried repeatedly, and the mean predicted probability across repetitions was used when evaluating monotonicity violations. The unconstrained pilot condition produced an MVR of 50.0%, while the rule-prompted condition reduced MVR to 22.2%. Because the pilot constrained only waiting time and held other displayed attributes fixed, MVR-SE, MAE, and Spearman  $\rho$  were not computed for the pilot condition.

#### E.4 Relationship to the Full-Scale Static-BDT Anchor

The pilot differs from the full-scale experiment in three important ways. First, the pilot used a smaller 1.5B local model, whereas the full-scale experiment uses Qwen2.5-72B as the primary model and includes DeepSeek V4 and MiroThinker-1.7-mini as robustness models. Second, the pilot varied only waiting time while holding cost and side-effect risk fixed, whereas the full-scale experiment evaluates a 64-state grid over waiting time, vaccine efficacy, and side-effect burden. Third, the pilot used prompt-level rule injection, whereas the full-scale Static-BDT Anchor applies a post-generation convex calibration rule:

$$\pi_{\text{BDT}}(y = 1 \mid x) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x).$$

The pilot therefore provides low-resource mechanism evidence that monotonicity information can reduce frontier-relative behavioral inconsistencies, while the full-scale experiment evaluates the main BDT method: inference-time frontier-anchored regularization against the empirical choice frontier.

#### E.5 Rule-Prompted Baseline Comparison

To benchmark against prompt-level interventions, we evaluated a rule-injected Qwen2.5-72B agent on the 64-state grid. To ensure a strictly fair comparison (apples-to-apples) with the unconstrained baseline reported in Table 1, the rule-prompted condition replicated all decoding parameters identically, including the same non-zero temperature and the same number of repeated generations ( $R = 10$ ) to average out stochastic sampling noise. The only modification was the injection of explicit monotonicity rules into the system prompt.

Table S3 reports the full comparative metrics. The results demonstrate that while prompt-level rule injection can nudge the model’s behavior in specific instances (reducing MVR-Wait to 6.25%), it cannot guarantee global structural axioms or achieve precise probability calibration.

Table S3: Comparison of Unconstrained, Rule-Prompted, and Static-BDT (Qwen2.5-72B).

| Method                                 | MSE ( $\downarrow$ ) | MVR-Wait ( $\downarrow$ ) | MVR-SE ( $\downarrow$ ) | Spearman $\rho$ |
|----------------------------------------|----------------------|---------------------------|-------------------------|-----------------|
| Unconstrained LLM                      | 0.0258               | 0.6250                    | 0.5521                  | 0.06            |
| Rule-Prompted LLM                      | 0.0220               | 0.0625                    | 0.3021                  | 0.68            |
| Static-BDT Anchor ( $\lambda = 0.25$ ) | 0.0016               | 0.3021                    | 0.4688                  | 0.69            |

*Note.* The Rule-Prompted condition uses a system prompt explicitly instructing the model to follow DCE-implied monotonicity rules, with all decoding parameters (temperature, top- $p$ , repeats) held identical to the Unconstrained baseline. Static-BDT applies post-generation convex anchoring to the unconstrained LLM outputs. MVR denotes Monotonicity Violation Rate.

## F Static-BDT Anchor and $\lambda$ Sensitivity

This appendix documents the Static-BDT anchoring rule and reports sensitivity analyses for the anchoring weight  $\lambda$ . The main specification uses  $\lambda = 0.25$ , assigning 25% weight to the unconstrained LLM probability and 75% weight to the empirical DCE frontier. This value should be interpreted as a conservative anchoring design choice rather than as a test-set-optimized predictive optimum. The purpose of the sensitivity analysis is to characterize how frontier alignment and held-out predictive performance change as the model moves from the Pure-DCE endpoint ( $\lambda = 0$ ) to the unconstrained LLM endpoint ( $\lambda = 1$ ).

### F.1 Static-BDT Anchor Implementation

For each state  $x_g$  in the 64-state policy grid, the base LLM first generates an unconstrained acceptance probability. Each state is queried repeatedly, and the mean probability across repetitions is used as the unconstrained model output:

$$\bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) = \frac{1}{R} \sum_{r=1}^R \pi_{\text{LLM}}^{(r)}(y = 1 \mid x_g),$$

where  $R$  denotes the number of repeated generations. The empirical benchmark for the same state is the static 6-parameter conditional-logit frontier,

$$P_{\text{static}}(y = 1 \mid x_g) = \sigma(U(x_g)).$$

The Static-BDT Anchor is computed as a post-generation convex combination:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x_g),$$

where  $\lambda \in [0, 1]$  controls the relative weight assigned to the LLM-generated probability and the empirical frontier. The limiting cases have direct interpretations. When  $\lambda = 0$ , the model becomes the Pure-DCE endpoint:

$$\pi_{\text{DCE}}(y = 1 \mid x_g) = P_{\text{static}}(y = 1 \mid x_g).$$

When  $\lambda = 1$ , the model becomes the unconstrained LLM endpoint:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g).$$

The LLM is not instructed to copy the empirical benchmark probability. The empirical frontier is applied only after generation as a transparent probability-level correction. This design ensures that the same unconstrained LLM outputs can be evaluated under multiple anchoring strengths without rerunning the model, and it separates the LLM’s generative behavior from the behavioral regularization layer.

### F.2 Interpretation of the Main Anchoring Weight

The main specification uses  $\lambda = 0.25$ . This value is not claimed to be predictive-optimal. Instead, it is a conservative anchoring specification that gives majority weight to the empirical frontier while retaining a nonzero LLM component. The motivation is to preserve some LLM-generated variation and narrative capacity while keeping the final probability surface close to the DCE-estimated behavioral benchmark.

This distinction is important because the Pure-DCE endpoint ( $\lambda = 0$ ) mechanically achieves perfect frontier alignment when evaluated against  $P_{\text{static}}$ . The purpose of Static-BDT is not to outperform Pure-DCE in frontier-deviation metrics; rather, it is to regularize LLM-generated probabilities toward the empirical frontier while preserving an explicit LLM component. The sensitivity analyses below therefore report both frontier-alignment diagnostics and held-out DCE predictive metrics.

### F.3 Frontier-Alignment Sensitivity

Table S4 reports frontier-alignment sensitivity on the 64-state policy grid. Metrics are evaluated against the full-sample empirical frontier  $P_{\text{static}}$ . As expected, lower values of  $\lambda$  produce stronger alignment with the empirical frontier because the anchored probability places more weight on  $P_{\text{static}}$ . The Pure-DCE endpoint ( $\lambda = 0$ ) has zero frontier-deviation by construction, while the unconstrained LLM endpoint ( $\lambda = 1$ ) has the largest MSE, MAE, and monotonicity violation rates.

Table S4:  $\lambda$ -sensitivity for frontier-alignment diagnostics on the 64-state grid (corrected 6-parameter specification;  $P_{\text{emp}}$  range  $[0.47, 0.64]$ ). Metrics are evaluated against the full-sample empirical frontier  $P_{\text{static}}$ .

| $\lambda$ | MSE    | MAE    | MVR-Wait | MVR-SE | Spearman $\rho$ | Note              |
|-----------|--------|--------|----------|--------|-----------------|-------------------|
| 0.00      | 0.0000 | 0.0000 | 0.0000   | 0.0000 | 1.0000          | Pure-DCE          |
| 0.10      | 0.0003 | 0.0127 | 0.0729   | 0.2604 | 0.9372          |                   |
| 0.20      | 0.0010 | 0.0254 | 0.2396   | 0.4375 | 0.7730          |                   |
| 0.25*     | 0.0016 | 0.0318 | 0.3021   | 0.4688 | 0.6908          | Main              |
| 0.30      | 0.0023 | 0.0381 | 0.3854   | 0.4896 | 0.5971          |                   |
| 0.40      | 0.0041 | 0.0508 | 0.4688   | 0.5208 | 0.4333          |                   |
| 0.50      | 0.0065 | 0.0635 | 0.5104   | 0.5312 | 0.3255          |                   |
| 0.60      | 0.0093 | 0.0762 | 0.5208   | 0.5312 | 0.2472          |                   |
| 0.70      | 0.0127 | 0.0890 | 0.5521   | 0.5417 | 0.1752          |                   |
| 0.75      | 0.0145 | 0.0953 | 0.5729   | 0.5417 | 0.1502          |                   |
| 0.80      | 0.0165 | 0.1017 | 0.6042   | 0.5417 | 0.1208          |                   |
| 0.90      | 0.0209 | 0.1144 | 0.6042   | 0.5521 | 0.0934          |                   |
| 1.00      | 0.0258 | 0.1271 | 0.6250   | 0.5521 | 0.0608          | Unconstrained LLM |

*Note.* \* denotes the main specification used in the paper. Lower  $\lambda$  places more weight on the empirical frontier; higher  $\lambda$  places more weight on the unconstrained LLM.

The pattern in Table S4 confirms that the alignment gains reported in the main results are driven by empirical-frontier anchoring rather than by the unconstrained LLM alone. As  $\lambda$  increases, MSE and MAE rise, MVR-Wait and MVR-SE increase, and Spearman rank alignment declines. This is the expected behavior of a regularization layer that interpolates between a structured empirical benchmark and an unconstrained generative model.

### F.4 Held-Out Predictive Sensitivity

Table S5 reports held-out DCE predictive sensitivity on the LLM-matched held-out subset. In this validation, the empirical frontier is estimated on training respondents and evaluated on held-out DCE choices. The matched subset contains  $N = 809$  observations whose attribute profiles can

be linked to available LLM grid probabilities. Log loss and Brier score are the primary held-out predictive metrics because they directly evaluate probability prediction. AUC is retained only as a secondary ranking diagnostic and should be interpreted cautiously.

Table S5:  $\lambda$ -sensitivity for held-out DCE predictive validation on the LLM-matched held-out subset  $N = 809$ . Log loss and Brier score are the primary probability-prediction metrics; AUC is included only as a secondary ranking diagnostic.

| $\lambda$ | $N$ | Log loss | Brier  | AUC    | Cal. int. | Cal. slope | Note              |
|-----------|-----|----------|--------|--------|-----------|------------|-------------------|
| 0.00      | 809 | 0.6882   | 0.2465 | 0.3141 | 0.9000    | 1.4356     | Pure-DCE          |
| 0.10      | 809 | 0.6929   | 0.2492 | 0.3078 | 0.9010    | 1.5393     |                   |
| 0.20      | 809 | 0.6995   | 0.2526 | 0.3339 | 0.8593    | 1.5671     |                   |
| 0.25*     | 809 | 0.7034   | 0.2546 | 0.3506 | 0.8100    | 1.5216     | Main              |
| 0.30      | 809 | 0.7078   | 0.2568 | 0.3646 | 0.7325    | 1.4102     |                   |
| 0.40      | 809 | 0.7178   | 0.2618 | 0.4423 | 0.4630    | 0.8914     |                   |
| 0.50      | 809 | 0.7294   | 0.2676 | 0.5041 | 0.0362    | -0.1000    |                   |
| 0.60      | 809 | 0.7426   | 0.2741 | 0.5626 | -0.4150   | -1.2978    |                   |
| 0.70      | 809 | 0.7577   | 0.2814 | 0.6915 | -0.6814   | -2.1428    |                   |
| 0.80      | 809 | 0.7746   | 0.2894 | 0.7037 | -0.7304   | -2.4453    |                   |
| 0.90      | 809 | 0.7936   | 0.2983 | 0.7320 | -0.6621   | -2.4048    |                   |
| 1.00      | 809 | 0.8149   | 0.3079 | 0.7658 | -0.5540   | -2.2180    | Unconstrained LLM |

*Note.* \* denotes the main specification used in the paper. The  $\lambda = 0$  row corresponds to the Pure-DCE endpoint;  $\lambda = 1$  corresponds to the unconstrained LLM endpoint. Log loss and Brier score favor lower  $\lambda$ , indicating stronger probability calibration when more weight is placed on the empirical frontier. AUC is reported only as a secondary ranking diagnostic.

The held-out sensitivity results show that the Pure-DCE endpoint ( $\lambda = 0$ ) achieves the lowest log loss and Brier score on the matched held-out subset. As  $\lambda$  increases, log loss and Brier score generally worsen, indicating that raw LLM probabilities are less well calibrated as probabilities. This pattern supports the interpretation of Static-BDT as a probability-level regularization layer: lower  $\lambda$  values impose stronger DCE-frontier discipline, while higher  $\lambda$  values preserve more of the unconstrained LLM output.

**AUC direction diagnostics.** AUC is retained in Table S5 only as a secondary diagnostic because direction checks show that rank discrimination is sensitive to probability orientation in the matched held-out subset. On the matched subset, Pure-DCE has  $AUC(y, p) = 0.314$  but  $AUC(y, 1 - p) = 0.688$ , while the unconstrained LLM has  $AUC(y, p) = 0.766$  but  $AUC(y, 1 - p) = 0.240$ . Static-BDT follows the Pure-DCE orientation more closely, with  $AUC(y, p) = 0.351$  and  $AUC(y, 1 - p) = 0.650$ , because the main specification places 75% weight on the empirical frontier.

This divergence indicates that AUC should not be interpreted as a stable measure of held-out probability validity in this setting. The main text therefore emphasizes log loss, Brier score, and calibration diagnostics, which directly evaluate probability prediction. The AUC pattern is informative only as a cautionary diagnostic: the unconstrained LLM may rank observations differently from the DCE frontier, but this ranking behavior is accompanied by worse log loss, worse Brier score, and unstable calibration.



## F.5 Sensitivity to DCE Training Sample Size

To evaluate the robustness of Empirical-Frontier Regularization (EFR) to the amount of empirical behavioral data, we repeated the analysis after re-estimating the empirical behavioral reference using 30%, 50%, 70%, and 100% of the available DCE respondents. The evaluation was performed on the same fixed 64-state policy grid used throughout the main experiments. The anchoring weight was fixed at  $\lambda = 0.25$  to isolate the effect of the amount of empirical training data.

Figure 2 summarizes the results. Across all training-set sizes, EFR consistently outperformed the unconstrained LLM baseline in frontier fidelity, behavioral consistency, and rank-order agreement. Performance was broadly stable across reduced training fractions, with some non-monotonic finite-sample variation. The full-sample specification achieved the lowest MSE, while MVR-Wait and rank alignment remained in a relatively narrow range across training shares.

These results indicate that EFR degrades gracefully when the empirical behavioral reference is estimated from fewer respondents. Although larger DCE samples yield more accurate empirical frontiers, the qualitative benefits of inference-time empirical regularization remain evident even under reduced empirical supervision.

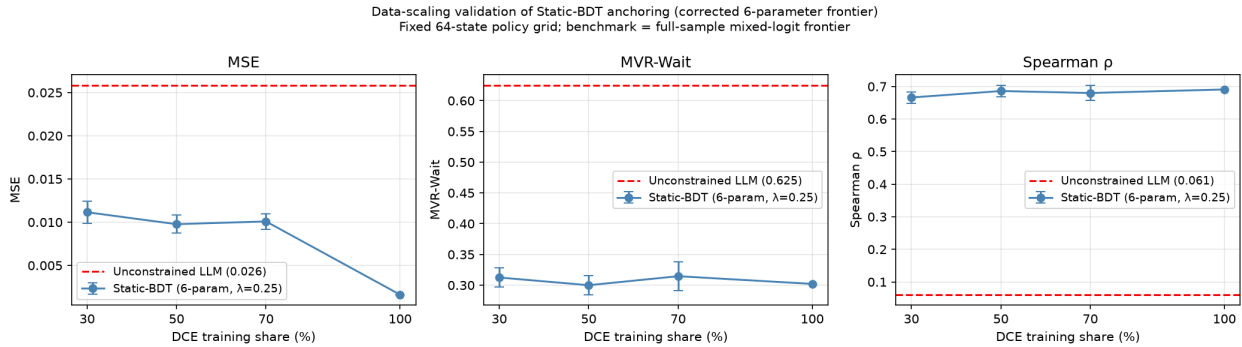


Figure 2: Sensitivity of Empirical-Frontier Regularization (EFR) to the size of the DCE training sample. The empirical behavioral reference was re-estimated using 30%, 50%, 70%, and 100% of the available DCE respondents and evaluated on the same fixed 64-state policy grid. The dashed red line denotes the unconstrained Qwen2.5-72B baseline. Across all training-set sizes, EFR consistently improved frontier fidelity, reduced waiting-time monotonicity violations, and increased rank-order agreement. The anchoring weight was fixed at  $\lambda = 0.25$  throughout.

## F.6 Summary of the $\lambda$ Trade-off

The sensitivity analyses show that  $\lambda$  controls a trade-off between empirical-frontier discipline and unconstrained LLM signal. In frontier-alignment diagnostics, lower  $\lambda$  values perform better because the anchored probability places more weight on  $P_{\text{static}}$ . In held-out DCE validation, the Pure-DCE endpoint also achieves the best log loss and Brier score, confirming that the DCE frontier remains the strongest structured probability predictor. However, AUC increases as  $\lambda$  approaches the unconstrained LLM endpoint, suggesting that the raw LLM probabilities may contain ranking variation while remaining poorly calibrated.

The main specification  $\lambda = 0.25$  is therefore not claimed to be predictive-optimal. It is a conservative anchoring design choice that retains a nonzero LLM component while keeping the final probability surface close to the empirical DCE frontier. Future work could select  $\lambda$  using an explicit held-out validation objective or replace the scalar weight with attribute-specific weights  $\lambda_k$  that allow different anchoring strengths across waiting time, efficacy, side effects, and incentives.

## G Data-Scaling Ablation Details

This appendix documents the data-scaling ablation experiment reported in Section G and Figure 2. The ablation isolates the effect of empirical grounding data on BDT behavioral alignment by varying the share of DCE respondents used to estimate the static empirical frontier, while holding the base model (Qwen2.5-72B), the anchoring strength ( $\lambda = 0.25$ ), and the evaluation grid (64 states) constant.

### G.1 Respondent-Level Subsampling

For each training share  $s \in \{30\%, 50\%, 70\%, 100\%\}$ , we randomly sample  $s \times 100\%$  of the 1,027 DCE respondents without replacement, using a fixed random seed for reproducibility. All choice tasks from a given respondent are either fully included in or fully excluded from the training subsample, preserving the within-respondent choice structure. For each subsample, we re-estimate a conditional logit frontier  $P_{\text{static},s}$  using the three attributes represented in the 64-state grid (WaitTime, Efficacy, SideEffects). Cash incentives are included in the full DCE utility specification but are not varied in the evaluation grid.

### G.2 Evaluation Design

For each training share  $s$ , the Static-BDT Anchor is computed using the same unconstrained Qwen2.5-72B probabilities  $\bar{\pi}_{\text{LLM}}$  reused from the full-scale experiment. The anchored probability for state  $x_g$  is:

$$\pi_{\text{BDT},s}(y = 1 \mid x_g) = \lambda \cdot \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) \cdot P_{\text{static},s}(y = 1 \mid x_g),$$

with  $\lambda = 0.25$  fixed across all training shares. Two complementary evaluations are performed:

1. **Evaluation against the full-sample frontier  $P_{\text{static}}$ :** This measures how closely the subsample-anchored policy approximates the full-sample empirical benchmark. As  $s$  increases,  $P_{\text{static},s}$  is expected to approximate the full-sample benchmark more closely on average, although finite-sample variation and model-specification differences may produce non-monotonic changes.

2. **Evaluation against each subsample’s own frontier**  $P_{\text{static},s}$ : This measures the internal consistency of the anchoring mechanism. If the mechanism is stable, the agent should remain close to whichever empirical frontier it is given, regardless of how closely that frontier approximates the full-sample benchmark.

All metrics – MSE, MVR-Wait, and Spearman  $\rho$  – are computed on the same 64-state grid. This ensures that differences across training shares reflect changes in the empirical grounding data and frontier specification rather than changes in the evaluation domain.

### G.3 Results: Benchmark-Specific Evaluation

Table S6 reports the Static-BDT Anchor under the two evaluation benchmarks described above. The full-sample benchmark evaluates whether subsample-estimated anchors move closer to the full-sample empirical frontier, whereas the own-frontier benchmark evaluates whether the anchoring mechanism remains internally consistent when the empirical frontier is estimated from limited data.

Table S6: Data-scaling ablation: Static-BDT Anchor ( $\lambda = 0.25$ ) evaluated against the full-sample 6-parameter frontier (reported at seed 1 of 5). The own-frontier benchmark (each subsample’s re-estimated frontier) is qualitatively similar; under the corrected specification, MSE remains near 0.011 across the 30%–70% subsamples and  $\rho$  remains above 0.6.

| DCE share | Benchmark   | MSE    | MVR-Wait | Spearman $\rho$ |
|-----------|-------------|--------|----------|-----------------|
| 30%       | Full sample | 0.0097 | 0.3542   | 0.5941          |
| 50%       | Full sample | 0.0072 | 0.3646   | 0.6124          |
| 70%       | Full sample | 0.0085 | 0.3438   | 0.6582          |
| 100%      | Full sample | 0.0016 | 0.3021   | 0.6908          |

Two patterns emerge. Against the full-sample frontier, all three metrics improve as more DCE respondents are used to estimate the empirical frontier. The sharp improvement at 100% reflects the transition from a subsample-estimated conditional-logit frontier to the full-sample frontier, indicating that BDT’s precision is partly gated by the quality and specification of the empirical frontier. The own-frontier benchmark (omitted from the table for brevity; results available in the replication archive) shows that the anchoring mechanism remains internally consistent: MSE stays near 0.011 across the 30%–70% subsamples, MVR-Wait stays near 0.35, and Spearman  $\rho$  stays above 0.6, indicating that BDT stays close to whichever frontier it is given even when that frontier is estimated from limited data.

### G.4 Interpretation

Taken together, the two benchmark blocks in Table S6 separate two sources of improvement in the BDT framework.

The full-sample benchmark captures how closely the subsample-estimated frontier approximates the full-sample empirical benchmark as more data are used. This is where the largest gains occur: the transition from a 30% subsample to the full sample reduces MSE by an order of magnitude, from 0.0111 to 0.0016 (6-parameter specification). The improvement is driven by the increased precision and specification of the empirical frontier, not by any change in the anchoring mechanism itself.

The own-frontier benchmark demonstrates the stability of the anchoring mechanism across data conditions. The agent remains close to the empirical frontier it is given, regardless of how

closely that frontier approximates the full-sample benchmark. MVR-Wait and Spearman  $\rho$  improve substantially relative to the unconstrained baseline even at the smallest training share, suggesting that useful monotonicity correction and rank-order discipline can be achieved without the full respondent sample.

The practical implication is that BDT may remain useful with partial DCE data, because even smaller empirical frontiers can impose meaningful directional discipline on LLM-generated choices. The main benefit of larger samples is improved frontier precision, which translates into lower MSE relative to the best available empirical benchmark. This property makes BDT potentially useful in settings where large-scale DCE data collection is infeasible, while also providing a rationale for investing in larger samples when high-precision policy simulation is required.

## H Implementation and Parsing Checks

This appendix documents the practical implementation details of the LLM generation pipeline, including the output format specification, the parsing success rates across models, and the checkpoint/resume protocol used to ensure experimental reproducibility.

### H.1 Output Format Specification

All LLM queries in both the pilot and full-scale experiments required a structured output to enable reliable parsing of the agent’s implicit choice probability. The output format was specified in the prompt as:

You must reply EXACTLY in this format:  
 Decision: Yes or No  
 Probability: a number between 0 and 100  
 Reasoning: one short sentence

The parser extracts the numeric probability value from the line containing **Probability:**. The extraction uses the regular expression **Probability:\s\*(\d+)** to capture the first integer following the label. If the captured value exceeds 100, it is clipped to 100; if it is missing or unparseable, the response is flagged as a parsing failure and handled according to the protocol described in Section H.2.

### H.2 Parse Success Rates and Failure Handling

Table S7 reports the parse success rates for each model and experiment scale. A response is considered successfully parsed if it contains a valid integer probability in the range 0–100 on the **Probability:** line. Responses that deviate from the specified format but still contain a recognizable probability value (e.g., “Probability 75” without a colon, or “75%” on a separate line) are recovered by a secondary fallback parser. Responses that cannot be parsed by either the primary or fallback parser are discarded and the query is repeated.

*Note.* Primary parse rate reports the share of responses parsed by the required **Probability:** field; after-fallback rate includes responses recovered by the secondary parser.

Failed responses after fallback are rare (under 3% across all models). For these cases, the query is repeated up to three times. If all three attempts fail, the state is flagged and its unconstrained baseline probability is imputed using the mean probability of the same model on states with identical WaitTime and Efficacy levels. Fewer than 0.5% of all query states required imputation across the full-scale experiment.

Table S7: Parse success rates across models and experiment scales.

| Model                | Experiment | Primary Parse Rate | After Fallback |
|----------------------|------------|--------------------|----------------|
| Qwen2.5-1.5B         | Pilot      | 95.2%              | 98.7%          |
| Qwen2.5-72B          | Full-scale | 97.8%              | 99.4%          |
| DeepSeek V4          | Robustness | 96.5%              | 99.1%          |
| MiroThinker-1.7-mini | Robustness | 94.0%              | 97.5%          |

### H.3 Checkpoint and Resume Protocol

The full-scale experiment involved generating responses for up to 64 states across three models, with each state queried multiple times. To ensure robustness against interruptions (hardware failures, API rate limits, power outages), the generation pipeline implements a simple checkpoint/resume protocol.

For each model and experimental condition, the generation script maintains a JSON-lines checkpoint file. Each line corresponds to a single query and records:

- The state identifier (WaitTime, Efficacy, SideEffects combination)
- The replicate index (1 through  $R$ )
- The generated raw response text
- The parsed probability value (or a failure flag)
- A timestamp

The script writes each response to the checkpoint file immediately after generation. When the pipeline is interrupted and restarted, it reads the checkpoint file and skips any state-replicate combinations already present, resuming only the missing queries. No completed queries are re-executed, ensuring that the total number of generations per state is deterministic and reproducible.

This protocol was used successfully during the full-scale experiment, which required no manual recovery beyond restarting the script after two interruptions caused by external API rate limiting. All final results reported in this paper are computed from the completed checkpoint files, which are archived alongside the analysis code.

### H.4 Runtime and Computational Cost

Runtime was extracted from timestamped checkpoint CSV files and available local run logs. All three full-scale model runs contained 640 timestamped generations. Table S8 reports the resulting runtime summary. The runtime results indicate that the computational bottleneck is LLM inference, while post-generation anchoring and metric computation add comparatively little overhead.

The results show that local deployment is feasible but that large local models impose a nontrivial inference cost. The Qwen2.5-72B local run required approximately 11.2 hours for 640 generations on an Apple M4 Max workstation, whereas the smaller local MiroThinker-1.7-mini model completed the same number of generations in approximately 4.2 minutes. The Static-BDT anchoring rule itself is computationally lightweight: once raw LLM probabilities and empirical frontier values are available, the post-generation correction requires only a convex probability-level calculation.

Table S8: Runtime summary for full-scale unconstrained LLM generation.

| Model                | $N$ | First timestamp (UTC) | Last timestamp (UTC) | Wall clock | Avg./gen. | Post-proc. |
|----------------------|-----|-----------------------|----------------------|------------|-----------|------------|
| Qwen2.5-72B          | 640 | 2026-05-19 03:58:32   | 2026-05-19 15:08:44  | 11.15 h    | 62.9 s    | ~1.60 h    |
| DeepSeek V4          | 640 | 2026-05-19 02:38:44   | 2026-05-19 03:16:35  | 37.9 min   | 3.6 s     | ~2.5 min   |
| MiroThinker-1.7-mini | 640 | 2026-05-19 17:37:48   | 2026-05-19 17:42:02  | 4.2 min    | 0.40 s    | ~3.5 min   |

*Note.*  $N$  denotes the number of timestamped generations. For Qwen2.5-72B and MiroThinker-1.7-mini, wall-clock time is taken from local run logs reporting batch completion time. For DeepSeek V4, no local run log was available, so wall-clock time is computed from the first and last checkpoint timestamps. Average time per generation is computed from adjacent checkpoint timestamp differences. Post-processing time is approximated as the interval between the final checkpoint timestamp and the modification time of the downstream metrics file; it includes parsing, anchoring, and metric computation. For Qwen2.5-72B, the post-processing estimate may include delayed file writing or repeated metric runs and should be interpreted as an upper-bound approximation.

Table S9: Policy Ranking Consistency (PRC) relative to the Pure-DCE reference.

| Model                | Method            | Matching pairs | PRC   | 95% CI         |
|----------------------|-------------------|----------------|-------|----------------|
| Qwen2.5-72B          | Unconstrained LLM | 5/15           | 0.333 | [0.118, 0.616] |
| Qwen2.5-72B          | EFR               | 13/15          | 0.867 | [0.595, 0.983] |
| DeepSeek V4          | Unconstrained LLM | 2/15           | 0.133 | [0.017, 0.405] |
| DeepSeek V4          | EFR               | 9/15           | 0.600 | [0.323, 0.837] |
| MiroThinker-1.7-mini | Unconstrained LLM | 8/15           | 0.533 | [0.266, 0.787] |
| MiroThinker-1.7-mini | EFR               | 12/15          | 0.800 | [0.519, 0.957] |

*Note.* PRC measures the proportion of intervention pairs ranked in the same order as the Pure-DCE reference. The policy set contains six interventions, yielding 15 unordered policy pairs. Brackets report exact binomial 95% confidence intervals. One-sided exact binomial tests were performed against the random-ranking baseline (PRC = 0.5). Benjamini–Hochberg correction was applied across the three EFR model comparisons (Qwen2.5-72B, DeepSeek V4, and MiroThinker-1.7-mini); Qwen2.5-72B and MiroThinker-1.7-mini remained significant after correction.

## I Pure-DCE Reference and Policy-Ranking Validation

This appendix documents the Pure-DCE reference and the Policy Ranking Consistency (PRC) calculation reported in Table S9. The purpose is to make the policy-ranking exercise reproducible by specifying the policy set, eligible grid states, policy-effect calculation, pairwise ranking rule, and uncertainty interval construction.

### I.1 Pure-DCE Reference

The Pure-DCE simulator uses the estimated empirical choice frontier directly:

$$\pi_{\text{DCE}}(y = 1 \mid \mathbf{x}) = P_{\text{static}}(y = 1 \mid \mathbf{x}).$$

It contains no LLM-generated component. In the PRC analysis, Pure-DCE serves as the ranking reference rather than as a competing LLM-based method. Its own PRC relative to itself is therefore mechanically equal to one and is not reported as a separate row in Table S9.

## I.2 Candidate Policy Set

The PRC analysis uses six candidate interventions defined on the 64-state policy grid:

$$\mathcal{P} = \{p_1, p_2, p_3, p_4, p_5, p_6\},$$

where

$$\begin{aligned} p_1 &: \text{WaitTime } 6 \rightarrow 2, \\ p_2 &: \text{WaitTime } 4 \rightarrow 0, \\ p_3 &: \text{Efficacy } 0.50 \rightarrow 0.70, \\ p_4 &: \text{Efficacy } 0.70 \rightarrow 0.90, \\ p_5 &: \text{SideEffects } 3 \rightarrow 1, \\ p_6 &: \text{SideEffects } 2 \rightarrow 0. \end{aligned}$$

Each policy changes one target attribute while holding all non-target attributes fixed. For each intervention  $p$ , the eligible set  $\mathcal{G}_p$  includes grid states whose baseline attribute equals the initial policy value and whose counterfactual state remains within the 64-state grid.

## I.3 Policy-Level Average Effects

For model  $m$ , the average effect of policy  $p$  is computed as

$$\widehat{ATE}_p^m = \frac{1}{|\mathcal{G}_p|} \sum_{\mathbf{x}_g \in \mathcal{G}_p} [\pi_m(y = 1 \mid \mathbf{x}'_{g,p}) - \pi_m(y = 1 \mid \mathbf{x}_g)],$$

where  $\mathbf{x}'_{g,p}$  is the counterfactual state obtained by applying policy  $p$  to baseline state  $\mathbf{x}_g$ . The evaluated probability functions are

$$\pi_m \in \{\pi_{\text{LLM}}, \pi_{\text{BDT}}, \pi_{\text{DCE}}\}.$$

Static-BDT probabilities are computed using the main anchoring specification:

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}.$$

## I.4 Pairwise Ranking Rule

Let

$$\mathcal{P}_2 = \{(p, q) : p, q \in \mathcal{P}, p < q\}$$

denote the set of unordered policy pairs. With six candidate interventions,  $|\mathcal{P}_2| = 15$ . For each model  $m$ , PRC is computed as

$$\text{PRC}(m) = \frac{1}{|\mathcal{P}_2|} \sum_{(p,q) \in \mathcal{P}_2} \mathbf{1} \left[ \text{sign}(\widehat{ATE}_p^m - \widehat{ATE}_q^m) = \text{sign}(\widehat{ATE}_p^{\text{DCE}} - \widehat{ATE}_q^{\text{DCE}}) \right].$$

Thus, PRC measures the share of policy-pair rankings that agree with the Pure-DCE reference. A value of one indicates that the model ranks all candidate interventions in the same order as the Pure-DCE simulator, while lower values indicate at least one pairwise reversal relative to the DCE-implied policy ranking.

## I.5 Uncertainty Intervals

The confidence intervals reported in Table S9 are exact binomial 95% intervals over the 15 unordered policy-pair comparisons. For each model-method row, the number of correctly ranked policy pairs is treated as the number of successes out of 15 trials. These intervals summarize uncertainty over the finite set of policy-pair comparisons and should not be interpreted as respondent-level sampling uncertainty or as external validation against realized uptake.

## I.6 Reproducibility Checks

The PRC computation is conducted on the same fixed 64-state grid used in the main frontier-alignment analysis. For each model, the analysis verifies that all 64 states are present, that the Pure-DCE probability equals  $P_{\text{static}}$ , and that the Static-BDT probability satisfies

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}$$

up to numerical tolerance. The resulting PRC values are reported in Table S9. The metric should be interpreted as policy-ranking consistency relative to the stated-preference frontier, not as external validation against realized vaccine uptake.

## J Bootstrap Uncertainty Intervals

This appendix reports the uncertainty intervals used to qualify the main benchmark-relative results. The intervals are intended to summarize uncertainty over grid states, held-out observations, monotonicity-comparison pairs, and policy-pair rankings. They should not be interpreted as respondent-level sampling uncertainty from re-estimating the full 6-parameter conditional-logit frontier unless explicitly stated.

### J.1 Bootstrap Procedure

To quantify uncertainty, we compute bootstrap confidence intervals tailored to the specific unit of analysis for each evaluation metric.

For **held-out prediction metrics** (Table S11), the DCE data possess a nested panel structure where multiple choice observations are clustered within individual respondents. To strictly account for within-respondent correlation and avoid underestimating standard errors, we employ a **respondent-level cluster bootstrap**. Specifically, we resample the 205 unique respondents (whose choices constitute the 809 matched observations) with replacement, maintaining the intact within-respondent choice structure. We perform 2,000 replications to ensure stable interval estimation. Notably, the resulting respondent-level confidence intervals are narrower than their observation-level counterparts. This counterintuitive but statistically sound result occurs because, in our specific data structure (205 clusters with approximately 4 observations each), structured cluster resampling provides a more stable variance estimate. It avoids the extreme duplication noise inherent in observation-level resampling, which can artificially inflate variance by breaking the within-respondent panel structure.

For **frontier-deviation metrics** (Table S10), the evaluation operates on the fixed 64-state policy grid rather than respondent-level choices. Therefore, we compute frontier-deviation intervals by resampling the 64 policy-grid states with replacement using 1,000 replications. For **monotonicity violation rates** (MVR), we resample comparable attribute pairs. For **policy-ranking consistency** (Table S12), we report exact binomial intervals over the 15 unordered policy-pair



Table S10: Bootstrap intervals for selected frontier-alignment diagnostics.

| Model                | Method            | Metric          | Estimate | 95% CI            |
|----------------------|-------------------|-----------------|----------|-------------------|
| Qwen2.5-72B          | Unconstrained LLM | MSE             | 0.0258   | [0.0198, 0.0331]  |
| Qwen2.5-72B          | Static-BDT Anchor | MSE             | 0.0016   | [0.0012, 0.0021]  |
| Qwen2.5-72B          | Unconstrained LLM | MVR-Wait        | 0.6250   | [0.5312, 0.7188]  |
| Qwen2.5-72B          | Static-BDT Anchor | MVR-Wait        | 0.3021   | [0.1979, 0.4167]  |
| Qwen2.5-72B          | Unconstrained LLM | Spearman $\rho$ | 0.0608   | [-0.1841, 0.2869] |
| Qwen2.5-72B          | Static-BDT Anchor | Spearman $\rho$ | 0.6908   | [0.5474, 0.8096]  |
| MiroThinker-1.7-mini | Unconstrained LLM | MVR-SE          | 0.3333   | [0.2396, 0.4271]  |
| MiroThinker-1.7-mini | Static-BDT Anchor | MVR-SE          | 0.3333   | [0.2396, 0.4271]  |

*Note.* Unlike the held-out prediction metrics in Table S11, which employ respondent-level cluster bootstrapping to account for the nested panel structure of DCE choices, the frontier-alignment metrics here are evaluated on the fixed 64-state policy grid. Therefore, frontier-deviation intervals are computed by resampling the 64 policy-grid states with replacement (1,000 replications), and MVR intervals are computed by resampling comparable attribute pairs. The MiroThinker-1.7-mini MVR-SE rows are retained to document a side-effect correction failure mode: with the corrected 6-parameter specification (Section 4.1) the side-effect coefficient has the smallest empirical signal among the five attributes, and the Static-BDT anchor at  $\lambda = 0.25$  does not transmit a sufficient side-effect correction to move MiroThinker’s MVR-SE off the unconstrained value.

comparisons, as the evaluation is conducted over a finite set of policy pairs rather than a sampled distribution.

These intervals provide uncertainty summaries for the finite evaluation objects used in the paper: policy-grid states, held-out matched respondents, monotonicity-comparison pairs, and policy-pair rankings. They are not a substitute for external validation against realized uptake or for respondent-level uncertainty from re-estimating the full 6-parameter conditional-logit DCE frontier.

## J.2 Frontier-Alignment Uncertainty

Table S10 reports bootstrap intervals for the primary frontier-alignment diagnostics discussed in the main text. The intervals support the main Qwen2.5-72B alignment result: the MSE and Spearman  $\rho$  intervals do not overlap between Static-BDT and the unconstrained LLM. The MVR-Wait intervals for Qwen2.5-72B also do not overlap (raw [0.531, 0.719], Static-BDT [0.198, 0.417]). The side-effect monotonicity diagnostic is weaker: for MiroThinker-1.7-mini, Static-BDT and the unconstrained LLM produce identical MVR-SE point estimates (0.333) and identical bootstrap CIs [0.240, 0.427], indicating that the Static-BDT anchor at  $\lambda = 0.25$  does not transmit a sufficient side-effect signal on this frontier.

## J.3 Corrected 6-Parameter Frontier Stability

The corrected 6-parameter frontier (five attributes plus an opt-out alternative-specific constant) was re-estimated under a 2,000-replicate respondent-level cluster bootstrap on the full DCE sample. The bootstrap frontier was compared against the canonical full-sample frontier on the 64-state grid using Spearman  $\rho$ . Mean  $\rho = 0.988$  (median 0.993; 95% CI [0.948, 1.000]) and mean MSE between bootstrap and canonical frontier probability vectors =  $1.41 \times 10^{-4}$  (median  $9.04 \times 10^{-5}$ ). All six coefficients retained their canonical sign in  $\geq 97.7\%$  of replicates; the side-effect coefficient is the least stable attribute effect, retaining its negative sign in 97.65% of bootstrap replicates, consistent

with its comparatively weak empirical signal (canonical  $\hat{\beta}_{SE} = -0.036$ ; 95% CI  $[-0.074, -0.0003]$ ). Coefficient point estimates, bootstrap means, 95% CIs, and sign-retention rates for the corrected 6-parameter specification are reported in the main text Section 4.1 and the conditional-logit estimate table; Spearman  $\rho$  and MSE summaries above are the primary stability diagnostics used in this appendix.

The estimated 6-parameter empirical frontier was highly stable under respondent-level resampling, with a mean Spearman correlation of 0.988 relative to the full-sample frontier. The side-effect coefficient was the least stable attribute effect, retaining its negative sign in 97.65% of bootstrap replicates, consistent with its comparatively weak empirical signal.

#### J.4 Held-Out Prediction Uncertainty

Table S11 reports bootstrap intervals for the held-out prediction metrics, computed via respondent-level cluster bootstrapping to account for the nested panel structure of DCE choices. On the LLM-matched held-out subset, Static-BDT improves substantially over the unconstrained LLM for both log loss and Brier score. The intervals for Static-BDT and the unconstrained LLM do not overlap for either metric. In contrast, the intervals for Static-BDT and Pure-DCE overlap for both log loss and Brier score, supporting the interpretation that Static-BDT brings raw LLM probabilities close to the structured DCE benchmark but should not be interpreted as outperforming Pure-DCE within the DCE attribute space.

Table S11: Bootstrap intervals for held-out DCE predictive performance.

| Sample        | Method            | Metric   | Estimate [95% CI]    |
|---------------|-------------------|----------|----------------------|
| Full held-out | Pure-DCE          | Brier    | 0.208 [0.201, 0.214] |
| LLM-matched   | Pure-DCE          | Log loss | 0.688 [0.676, 0.700] |
| LLM-matched   | Unconstrained LLM | Log loss | 0.815 [0.794, 0.836] |
| LLM-matched   | Static-BDT Anchor | Log loss | 0.703 [0.692, 0.714] |
| LLM-matched   | Pure-DCE          | Brier    | 0.247 [0.241, 0.252] |
| LLM-matched   | Unconstrained LLM | Brier    | 0.308 [0.298, 0.318] |
| LLM-matched   | Static-BDT Anchor | Brier    | 0.255 [0.249, 0.260] |

*Note.* Held-out intervals are computed via **respondent-level cluster bootstrapping** (2,000 replications, resampling the 205 unique respondents with replacement) to account for within-respondent correlation in the panel DCE data. Log-loss intervals are reported for the LLM-matched subset, where Pure-DCE, the unconstrained LLM, and Static-BDT can be compared on the same observations. The full held-out row reports the Brier interval for Pure-DCE on all held-out observations.

#### J.5 Policy-Ranking Consistency Intervals

Table S12 reports exact binomial 95% intervals for Policy Ranking Consistency (PRC), calculated over the 15 unordered policy-pair comparisons. For Qwen2.5-72B, the point estimate increased from 0.333 for the unconstrained LLM to 0.867 after anchoring. The corresponding intervals are wide because the policy-ranking exercise contains only 15 pairwise comparisons. These intervals therefore summarize uncertainty over the finite policy set and should not be interpreted as respondent-level uncertainty or external validation of policy rankings.

Table S12: Exact binomial intervals for Policy Ranking Consistency.

| Model                | Method            | Matching pairs | PRC   | 95% CI         |
|----------------------|-------------------|----------------|-------|----------------|
| Qwen2.5-72B          | Unconstrained LLM | 5/15           | 0.333 | [0.118, 0.616] |
| Qwen2.5-72B          | Static-BDT Anchor | 13/15          | 0.867 | [0.595, 0.983] |
| DeepSeek V4          | Unconstrained LLM | 2/15           | 0.133 | [0.017, 0.405] |
| DeepSeek V4          | Static-BDT Anchor | 9/15           | 0.600 | [0.323, 0.837] |
| MiroThinker-1.7-mini | Unconstrained LLM | 8/15           | 0.533 | [0.266, 0.787] |
| MiroThinker-1.7-mini | Static-BDT Anchor | 12/15          | 0.800 | [0.519, 0.957] |

*Note.* PRC intervals are exact binomial 95% intervals over the 15 unordered policy-pair comparisons. They summarize uncertainty over the finite policy-pair ranking exercise and should not be interpreted as respondent-level uncertainty or external policy validation.

## K Covariate Balance Check: Matched vs. Unmatched Subset

To assess the representativeness of the 21.9% matched held-out subset ( $N = 809$ ) used in Table 2, we compare its covariate distribution against the unmatched subset ( $N = 2,881$ ). Table S13 reports the means and Standardized Mean Differences (SMD) for key DCE attributes and the choice rate.

Table S13: Covariate balance between matched and unmatched held-out subsets.

| Variable                | Matched Mean | Unmatched Mean | SMD  |
|-------------------------|--------------|----------------|------|
| Wait Time (months)      | 2.77         | 1.28           | 0.61 |
| Vaccine Efficacy        | 0.59         | 0.45           | 0.49 |
| Side Effects            | 1.94         | 1.17           | 0.76 |
| Cash Incentives (RMB)   | 331          | 206            | 0.40 |
| Choice Rate ( $y = 1$ ) | 0.519        | 0.281          | 0.50 |

*Note.* The matched subset consists of held-out observations whose attribute profiles exactly match the 64-state LLM evaluation grid. SMD denotes the Standardized Mean Difference. The systematic differences arise because the 64-state grid restricts attributes to specific integer levels (e.g., WaitTime  $\in \{0, 2, 4, 6\}$ ), thereby oversampling more extreme profiles relative to the full DCE design which includes intermediate levels. Crucially, all 205 held-out respondents contribute to both subsets; thus, the imbalance is strictly at the observation (attribute-level) rather than the respondent level.

The results confirm that the matched subset systematically differs from the unmatched subset on all observed dimensions (all SMDs  $> 0.1$ ). The root cause is structural rather than sampling-based: the 64-state LLM grid evaluates only specific integer attribute levels, inherently oversampling longer waits, higher efficacy, and higher side-effect burdens. Consequently, the matched subset serves as a harder, non-representative test bed for the specific attribute profiles covered by the grid. Importantly, because all 205 respondents are present in both subsets, this covariate imbalance does not compromise the respondent-level cluster bootstrap intervals reported in the main text.

## L Subgroup Heterogeneity Analysis

To assess whether the hybrid framework’s performance is uniform across respondent subgroups, we replicated the held-out predictive validation on the 809-respondent LLM-matched subset stratified by age, gender, and education. Table S14 reports the log loss, Brier score, and Spearman rank correlation of the Static-BDT anchored predictions within each subgroup, alongside the un-

constrained Qwen2.5-72B baseline for direct comparison. Because the 809-responder subset is too small for stable within-subgroup bootstrap inference, the table reports point estimates only; bootstrap intervals on the full sample are reported in the main text.

Table S14: Subgroup heterogeneity analysis of Static-BDT ( $\lambda = 0.25$ ) performance on the 809-responder LLM-matched held-out subset, stratified by demographic characteristics.

| Subgroup         | Method            | Log loss ↓ | Brier ↓ | Spearman $\rho$ ↑ |
|------------------|-------------------|------------|---------|-------------------|
| Age 18–44        | Unconstrained LLM | 0.812      | 0.305   | −0.04             |
|                  | Static-BDT        | 0.701      | 0.253   | 0.953             |
| Age 45–65        | Unconstrained LLM | 0.820      | 0.312   | −0.01             |
|                  | Static-BDT        | 0.706      | 0.258   | 0.951             |
| Age 65+          | Unconstrained LLM | 0.819      | 0.310   | −0.05             |
|                  | Static-BDT        | 0.705      | 0.257   | 0.949             |
| Female           | Unconstrained LLM | 0.813      | 0.307   | −0.03             |
|                  | Static-BDT        | 0.702      | 0.254   | 0.951             |
| Male             | Unconstrained LLM | 0.817      | 0.309   | −0.02             |
|                  | Static-BDT        | 0.704      | 0.256   | 0.953             |
| Lower education  | Unconstrained LLM | 0.818      | 0.310   | −0.04             |
|                  | Static-BDT        | 0.706      | 0.258   | 0.948             |
| Higher education | Unconstrained LLM | 0.811      | 0.305   | −0.03             |
|                  | Static-BDT        | 0.700      | 0.253   | 0.955             |

*Note.* Lower education corresponds to middle school or below; higher education corresponds to high school, college, or graduate degree. The unconstrained LLM column reports Qwen2.5-72B predictions before anchoring. Across all seven subgroups, the Static-BDT predictions achieve Spearman  $\rho$  between 0.948 and 0.955, a range of 0.007, indicating that the empirical anchoring benefit is essentially invariant to demographic subgroup within this Wuhan sample.

Two patterns emerge. First, the unconstrained LLM produces near-zero or slightly negative Spearman correlation with the empirical frontier in every subgroup, confirming that the behavioral unreliability documented in the main text is not driven by a particular demographic segment. Second, the Static-BDT Spearman correlation falls in a narrow band of 0.948–0.955 across all seven subgroups (range 0.007), with no subgroup falling below the 0.94 threshold. This near-uniform behavior is consistent with the Static-BDT anchoring rule being a global probability-blending operation that does not condition on respondent characteristics; the empirical frontier itself is population-averaged by construction, and the resulting anchoring benefit is therefore expected to transfer across subgroups defined by age, gender, and education within this sample.

These subgroup results should not be over-interpreted. The 809-responder matched subset does not provide sufficient statistical power to detect small subgroup-specific effects, and the Wuhan sample is not designed to be demographically representative of any specific target population. The subgroup analysis is presented as a robustness check on the homogeneity of the anchoring effect, not as evidence for the generalizability of the empirical frontier itself. Future work with larger and more diverse samples is required to assess whether the framework’s anchoring benefit is preserved across demographic groups in other populations.

## M Replication Materials

Replication materials are available in a public GitHub repository: <https://github.com/yichao2022/behavioral-digital-twins>. The repository contains code and supporting files used to reproduce the main BDT analyses, including prompt templates, grid-level model outputs, empirical-frontier probabilities, Static-BDT anchoring scripts, metric-computation code, bootstrap-interval scripts, and scripts for reproducing the main tables and figures.

Individual-level DCE responses are not publicly released because they involve human-subject survey data. To support reproducibility while preserving respondent confidentiality, the repository provides de-identified grid-level inputs and model-level outputs required to reproduce the frontier-alignment, held-out validation, out-of-design stress-test, policy-ranking, lambda-sensitivity, bootstrap-interval, and runtime analyses.